# NORTH BENGAL INTERNATIONAL UNIVERSITY

## Lab Report : 1

**TOP** : CODE FOR IF-ELSE, NESTED IF, IF-ELSE-IF LADDER, SWITCH, NESTED FOR LOOP, NESTED WHILE LOOP, NESTED DO WHILE LOOP IN C++

**COURSE CODE** : CSE 0612-2122

**COURSE TITLE** : OBJECT ORIENTED PROGRAMMING SESSIONAL.


Submitted By:

**MD. TANVIR RAHMAN PRANTO**

**0712340012101005**

**CSE -26**

Submitted To:

**AMIT KUMAR PAUL**

**LECTURER, CSE**

## Code for if-else, Nested if, if-else-if ladder, switch, Nested for loop, Nested while loop, do while loop in c++

## # if-else:

**Code:**

```
#include <iostream>

int main() {
    int number;
    std::cout << "Enter a number: ";
    std::cin >> number;

        if (number > 10) {
        std::cout << "The number is greater than 10." << std::endl;
    } else {
        std::cout << "The number is 10 or less." << std::endl;
    }

    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

Enter a number: 05
The number is 10 or less.
PS C:\Users\Lenovo\Documents\c++>
```

## # Nested if:

**Code:**

```
#include <iostream>

int main() {
    int number;
    std::cout << "Enter a number: ";
    std::cin >> number;

    if (number > 0) {
        std::cout << "The number is positive." << std::endl;
        if (number % 2 == 0) {
            std::cout << "It is even." << std::endl;
        } else {
            std::cout << "It is odd." << std::endl;
        }
    } else if (number < 0) {
        std::cout << "The number is negative." << std::endl;
    } else {
        std::cout << "The number is zero." << std::endl;
    }
    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

Enter a number: 05
The number is positive.
It is odd.
PS C:\Users\Lenovo\Documents\c++>
```

## # if-else-if ladder:

**Code:**

```cpp
#include <iostream>

int main() {
    int age;
    std::cout << "Enter your age: ";
    std::cin >> age;

    // If-else-if ladder
    if (age >= 65) {
        std::cout << "You are a senior citizen." << std::endl;
    } else if (age >= 18) {
        std::cout << "You are an adult." << std::endl;
    } else if (age >= 13) {
        std::cout << "You are a teenager." << std::endl;
    } else {
        std::cout << "You are a child." << std::endl;
    }

    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

Enter your age: 25
You are an adult.
PS C:\Users\Lenovo\Documents\c++>
```

## # Switch:

**Code:**

```cpp
#include <iostream>

int main() {
    char grade;
    std::cout << "Enter your grade (A, B, C, D, F): ";
    std::cin >> grade;
```

```
    switch (grade) {
        case 'A':
            std::cout << "Excellent!" << std::endl;
            break;
        case 'B':
            std::cout << "Good job!" << std::endl;
            break;
        case 'C':
            std::cout << "Well done!" << std::endl;
            break;
        case 'D':
            std::cout << "You passed." << std::endl;
            break;
        case 'F':
            std::cout << "Better luck next time." << std::endl;
            break;
        default:
            std::cout << "Invalid grade!" << std::endl;
            break;
    }

    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

Enter your grade (A, B, C, D, F): A
Excellent!
PS C:\Users\Lenovo\Documents\c++>
```

## # Nested for loop:

**Code:**

```
#include <iostream>

int main() {

    int n = 5;
    for (int i = 1; i <= n; ++i) {
        for (int j = 1; j <= i; ++j) {
            std::cout << "* ";
        }
        std::cout << std::endl;
    }

    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

*
* *
* * *
* * * *
* * * * *
PS C:\Users\Lenovo\Documents\c++>
```

## # Nested While loop:

**Code:**

```
#include <iostream>

int main() {
    int i = 1;
    int j;

    while (i <= 5) {
        j = 1;
        while (j <= i) {
            std::cout << j << " ";
            ++j;
        }
        std::cout << std::endl;
        ++i;
    }

    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
PS C:\Users\Lenovo\Documents\c++>
```

## # Nested While loop:

**Code:**

```
#include <iostream>
int main() {
    int row = 1;
    int col;

    do {
        col = 1;
        do {
            std::cout << row * col << "\t";
            ++col;
        } while (col <= 5);
        std::cout << std::endl;
        ++row;
    } while (row <= 5);
    return 0;
}
```

**Output:**

```
OUTPUT    TERMINAL

1       2       3       4       5
2       4       6       8       10
3       6       9       12      15
4       8       12      16      20
5       10      15      20      25
PS C:\Users\Lenovo\Documents\c++>
```